



02451 Introduction to Machine Learning

Week 3: Principal Component Analysis

Bjørn Sand Jensen

17 February 2026

DTU Compute, Technical University of Denmark

Today

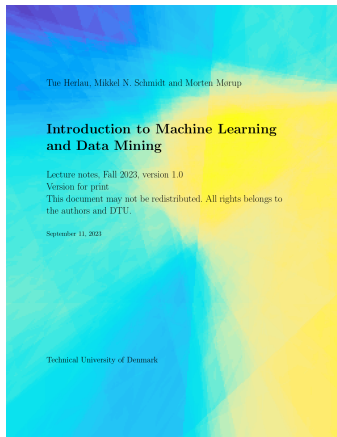
Feedback Groups of the day:

Nikoline Grete Aarø-Hansen, Louis Agbesi, Ella Fan Randlev Alsøe, Ghaith Bassem Altayeb, Kasper Hestbæk Bankler, Serif Catalgöl, Jamal Khaled Charkie, Yi Chen, Ekrem Degirmenci, Findlay Eldridge, Toms Evertovskis, Halfdan Elvis Fridal-Omeragic, Mikkel Vinther Fritzel, Greta Gakaite, Frederik Møgelvang Hansen, Mathias Højland, Christian Hussing, Markus Kühne Jakobsen, Joshua Bælum Jefferson, Jonathan Rahbek Kragh, Frederik Krøis, Jonas Krüger, Frederik Bloch Larsen, Marie Kirstine Lorenzen, Carl Johan von Lowzow, Lasse Lundbæk, Anne Mette Godballe Lundberg, Diego Medina Castello, Jesper Norborg, Anders Nørgaard, Sarah Notto, Emilie Nyholm-Christensen, August Witte Feentved Ødum, Alexander Frederik Olesen, Ronja Amalie Østergaard, Jakob Milthers Petersen, Alexander Anker Pieppgrass, Julius Rantanen, Mattia Russo, Emil Thorsen Staugaard, Stefan Artur Stefirta, Oscar Flynn Sunderland, Vincent Thisted Vaupel, Vasco Caetano Paliotes Coelho Veiga,

Reading/homework material:

Chapter 3

P3.1, P3.2



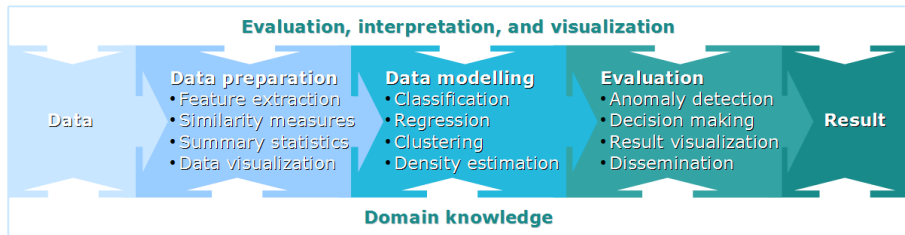
Lecture Schedule

- 1 Introduction, data, and visualization
3 February: C1, C2, C7
- 2 Summary statistics, similarity, and nearest neighbors
10 February: C4, C12
- 3 **Computational linear algebra and PCA**
17 February: C3
- 4 Probability theory, Bayes and Naive Bayes
24 February: C5, C6, C13 (Assignment 1-3 due on 26 February)
- 5 Bayesian inference and linear models
3 March: C5, C6, C8
- 6 Decision trees, overfitting and cross-validation
10 March: C9, C10
- 7 Performance evaluation and AUC
17 March: C11, C16 (Assignment 4-6 due on 19 March)
- 8 Artificial Neural Networks and Optimization
24 March: C15
- 9 Bias/Variance and ensemble methods
7 April: C14, 17
- 10 K-means and hierarchical clustering
14 April: C18 (Assignment 7-9 due on 16 April)
- 11 Mixture models and density estimation
21 April: C19, C20
- 12 Representation learning
28 April: Material provided separately
- 13 Recap and discussion of the exam
5 May: C1-C20 (Assignment 10-12 due on 7 May)

Online help: Piazza

Videos of lectures: <https://panopto.dtu.dk>
Streaming of lectures: Zoom (link on DTU Learn)

Learning Objectives



Learning Objectives

- Understand and apply matrix operations on data matrices
- Understand and apply principal component analysis for data visualization and feature extraction

Plan for today:

- Lecture 3 (13:00 – ~15:00)
 - Linear algebra revisited
 - Principal Component Analysis
 - Applications of PCA
- Exercises (15:00 – 17:00)

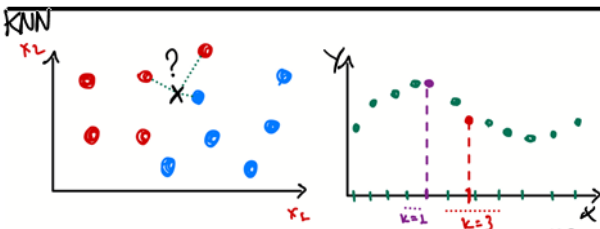
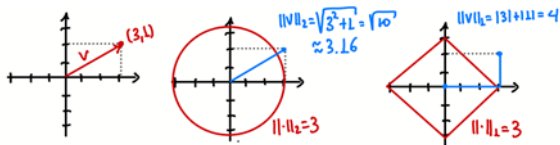
Practicalities and announcements

- Bug in exercise03.zip $\Rightarrow$ Use the new one uploaded today!
- Assignments: **Deadline for week 1-3 next week** - submit both HTML and ipynb!
- Issues/stuck with an assignment - use your TA (or Piazza)
- Homework problems solutions?
- The TAs will go over one of the homework problems or old exam questions at the end of sessions (around 16:45)

Recap lecture 2

- Summary statistics
- Vector space and distances
- k-Nearest Neighbors (kNN)

$$X = \begin{bmatrix} 1.1 & -4 \\ -2.1 & 3 \\ 4.3 & 4.3 \\ -0.1 & -1.1 \\ \vdots & \vdots \\ -2.1 & 5.5 \\ \hat{\mu}_{x_1} & \hat{\mu}_{x_2} \\ \hat{\sigma}_{x_1} & \hat{\sigma}_{x_2} \end{bmatrix}$$

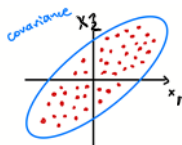


$$\hat{\mu}_x = \frac{1}{N} \sum_{i=1}^N x_i \quad \hat{\sigma}_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \hat{\mu}_x)^2$$

Covariance matrix $\hat{S}$

$$\text{Cov}[x^{(u)}, x^{(m)}] = \frac{1}{N-1} \sum_{i=1}^N (x_i^{(u)} - \hat{\mu}_{x^{(u)}})(x_i^{(m)} - \hat{\mu}_{x^{(m)}})$$

$$\hat{S} = \begin{bmatrix} \text{Cov}[x_1, x_1] & \text{Cov}[x_1, x_2] & \text{Cov}[x_1, x_3] \\ \vdots & \text{Cov}[x_2, x_2] & \text{Cov}[x_2, x_3] \\ \vdots & \vdots & \text{Cov}[x_3, x_3] \end{bmatrix}$$



Hint: Compute $\hat{S}$ using matrix notation

- Center the data: $\tilde{x}_i = x_i^{(u)} - \hat{\mu}_{x^{(u)}}$
- $\hat{S} = \frac{1}{N-1} \sum_{i=1}^N \tilde{x}_i \tilde{x}_i^T = \frac{1}{N-1} \tilde{X}^T \tilde{X} \quad (M \times M)$

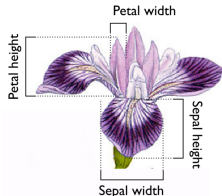
Recall: Data points as vectors

A data point is a set of **real** numbers represented as a vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_M \end{bmatrix} \in \mathbb{R}^M \quad \text{e.g. } \mathbf{x} = \begin{bmatrix} 5.1 \\ 3.5 \\ 1.4 \\ 0.2 \end{bmatrix}$$

A matrix is used to hold multiple data points (as rows)

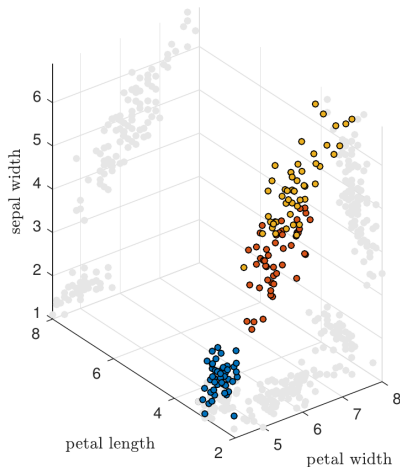
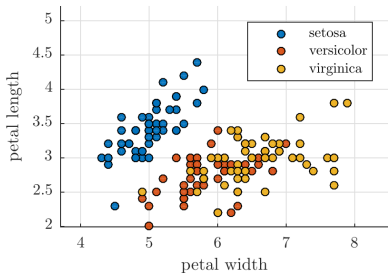
$$\mathbf{X} = \begin{bmatrix} x_{1,1} & \dots & x_{1,M} \\ \vdots & & \vdots \\ x_{N,1} & \dots & x_{N,M} \end{bmatrix} \in \mathbb{R}^{N \times M}$$



Sepal Length	Sepal Width	Petal Length	Petal Width
5.1	3.5	1.4	0.2
4.9	3.0	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2
5.8	2.7	5.1	1.9
$\vdots$	$\vdots$	$\vdots$	$\vdots$
5.7	2.8	4.1	1.3

$$\mathbf{X} = \begin{bmatrix} 5.1 & 3.5 & 1.4 & 0.2 \\ 4.9 & 3.0 & 1.4 & 0.2 \\ 4.7 & 3.2 & 1.3 & 0.2 \\ 4.6 & 3.1 & 1.5 & 0.2 \\ 5.8 & 2.7 & 5.1 & 1.9 \\ \vdots & \vdots & \vdots & \vdots \\ 5.7 & 2.8 & 4.1 & 1.3 \end{bmatrix}$$

Example: Visualization of high-dimensional data



Linear algebra for machine learning

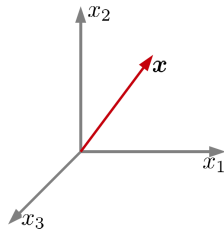
Data representation Data in ML is often represented as vectors and matrices.

Transformations Operations like scaling, rotations, and projections are naturally and efficiently formulated using matrix operations.

Dimensionality reduction and visualization Data is high-dimensional and sometimes requires reduction in dimensionality to find patterns and extract new features (e.g., as combinations of existing ones). Techniques like PCA use eigenvectors and eigenvalues.

Model formulation Functions are easily and compactly formulated in vector/matrix notation (e.g., linear models $f(\mathbf{x}; \mathbf{w}) = \mathbf{w}^\top \tilde{\mathbf{x}}$ and neural networks)

Optimization Modern ML models relies heavily on gradient methods. Gradients are conveniently represented as vectors and matrices (e.g. a Jacobian or Hessian)



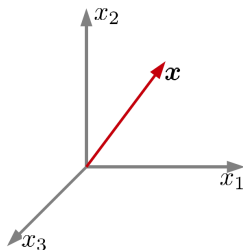
Vector spaces

- A M -dimensional vector space is just $\mathbb{R}^M$
- This is the set of all M -dimensional vectors
- A vector space is closed under linear combinations

$$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$$

$$a_1, a_2, \dots, a_n$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_M \end{bmatrix}$$



Distances and norms

- The (Euclidian) norm of a vector measures it's length (magnitude):

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{n=1}^N |x_n|^2} = \sqrt{\mathbf{x}^\top \mathbf{x}}$$

- The Frobenius norm of a matrix measures it's magnitude:

$$\|\mathbf{X}\|_F^2 = \sum_{i,j} x_{i,j}^2 = \text{trace}(\mathbf{X}\mathbf{X}^T) = \text{trace}(\mathbf{X}^T\mathbf{X})$$

Where trace takes the sum of the diagonal elements, i.e. $\text{trace}(\mathbf{A}) = \sum_{i=1}^N a_{i,i}$

Scaling, rotation and shear

Rotation:

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$x' = Rx$$

E.g. $R = \begin{bmatrix} 0.87 & -0.5 \\ 0.5 & 0.87 \end{bmatrix}$

Scaling and stretching:

$$S(\alpha_x, \alpha_y) = \begin{bmatrix} \alpha_x & 0 \\ 0 & \alpha_y \end{bmatrix}$$

$$x' = Sx$$

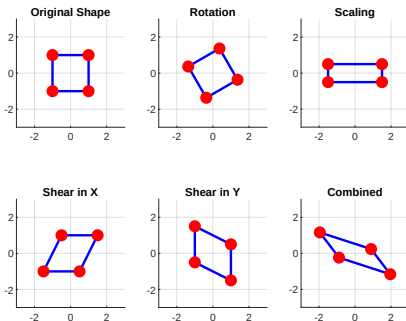
E.g. $S = \begin{bmatrix} 1.5 & 0 \\ 0 & 0.5 \end{bmatrix}$

Shear:

$$D_x(\alpha_x, \alpha_y, \beta) = \begin{bmatrix} \alpha_x & \beta \\ 0 & \alpha_y \end{bmatrix}$$

$$x' = D_x x$$

E.g. $D_x = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$



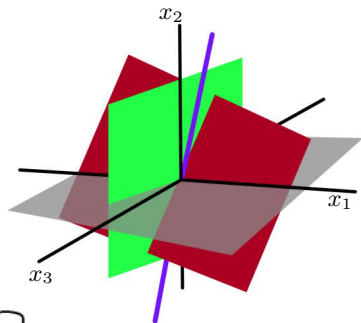
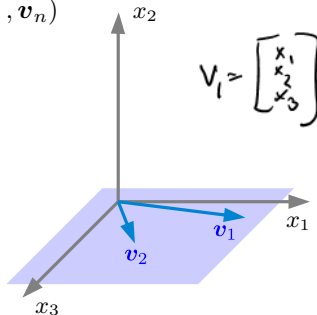
Subspaces

- A **subspace** generalizes the concept of a line/plane
- If we consider n vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ the span is then all linear combinations

$$\mathbf{z} = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n$$

and is said to be a subspace

$$V = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$$



Basis of a(sub)space

- Vectors $v_1, v_2, \dots, v_n$ are said to be **linearly independent** if

$$0 = a_1 v_1 + a_2 v_2 + \dots + a_n v_n$$

implies $a_1 = a_2 = \dots = a_n = 0$

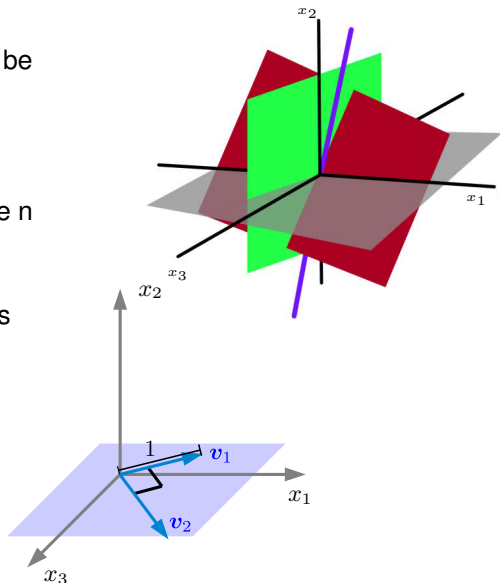
- A **basis** for a vector space, V , are n linearly independent vectors

$$V = \text{span}(v_1, v_2, \dots, v_n)$$

- A basis is orthonormal if the basis is orthogonal and of unit length

$$v_i^\top v_j = 0 \text{ for } i \neq j$$

$$\|v_i\| = 1$$

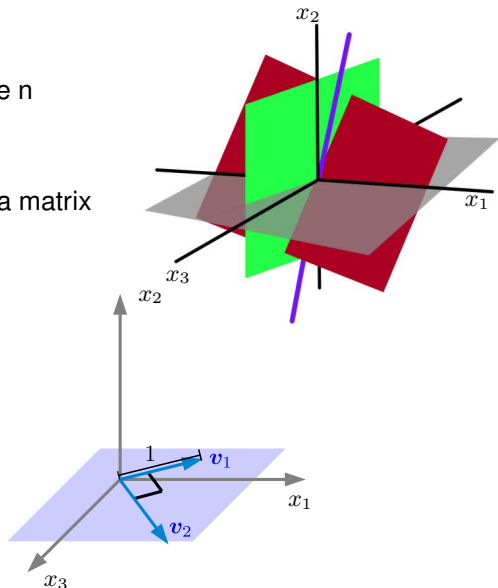


Basis of a (sub)space

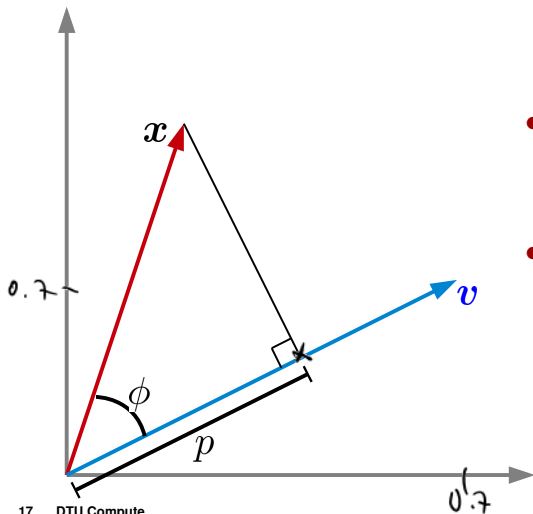
- A **basis** for a vector space, V , are n linearly independent vectors
 $V = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$
- We collect the basis vectors into a matrix

$$V = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & \cdots & | \end{bmatrix}$$

- If the basis is orthonormal, the matrix satisfies
 $V^\top V = I$ and $V^\top = V^{-1}$



Projection



- Angle between vectors

$$\cos(\phi) = \frac{v^\top x}{\|x\|_2 \|v\|_2}$$

- Length of projection

$$p = \|x\|_2 \cos(\phi) = \frac{v^\top x}{\|v\|_2}$$

- Projection onto the unit vector

$$p = v^\top x$$
$$= [0.7 \ 0.7] \begin{bmatrix} 0.5 \\ 1.1 \end{bmatrix}$$

Projection onto a subspace

- Projection onto a subspace**

- Subspace of dimension D defined by an orthonormal basis matrix V .
- Projection of x (M dimensional) onto V is given by

$$b^T = x^T V = [x_1, x_2, x_3] \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

- 'Reconstruction' can be found as:

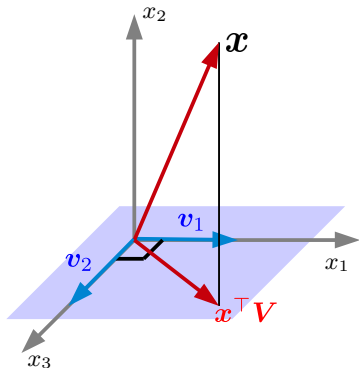
$$x' = Vb$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ x_3 \end{bmatrix}$$

Example I:

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$x^T V = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = [x_1 \quad x_3]$$



Projection onto a subspace

- **Projection onto a subspace**

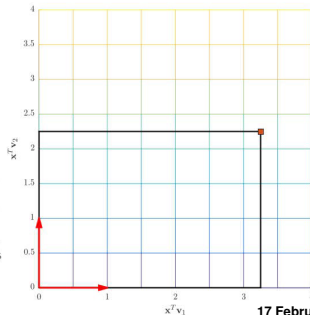
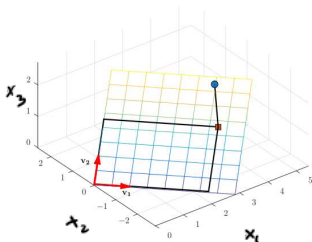
- Subspace of dimension D defined by an orthonormal basis matrix V .
- Projection of x (M dimensional) onto V given by

$$b^\top = x^\top V$$

- 'Reconstruction' can be found as:

$$x' = Vb$$

Example II:



Eigenvectors and eigenvalues

14:05

- Suppose

$$Av = \lambda v, \quad A \text{ is a } N \times N \text{ matrix}$$

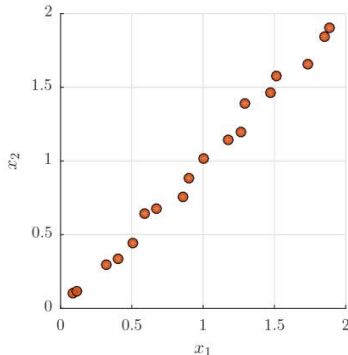
- We say v is an eigenvector with eigenvalue λ
- If A is symmetric: $A = A^T$ then A has orthogonal eigenvectors and eigenvalues that are real.

$$A = Q \Lambda Q^T$$

$$Q = \begin{bmatrix} | & | & | & \dots & | \\ v_1 & v_2 & v_3 & \dots & v_M \\ | & | & | & \dots & | \end{bmatrix} \quad \Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_M \end{bmatrix}$$

PCA for high-dimensional data

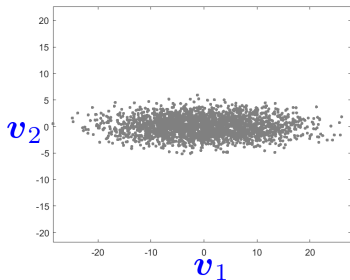
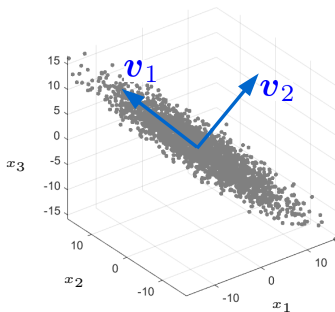
- Much data is high-dimensional
- We want to find a **lower**-dimensional representation of the **high**-dimensional data.



(2 dimensional but really (close to being) 1 dimensional)

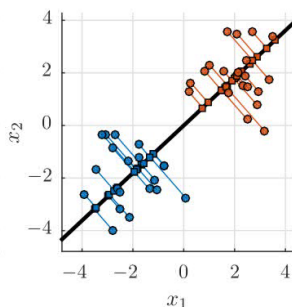
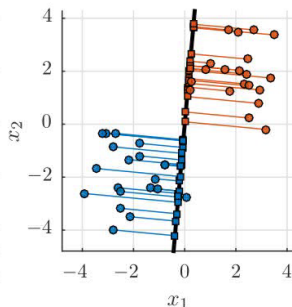
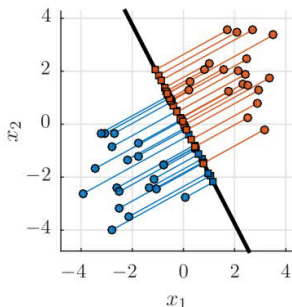
PCA for high-dimensional data

- Much data is high-dimensional
- We can project **high**-dimensional data to a **lower**-dimensional subspace.
- But what is a good projection?



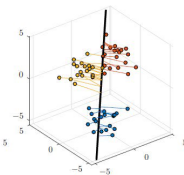
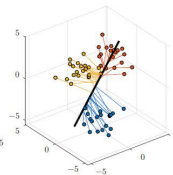
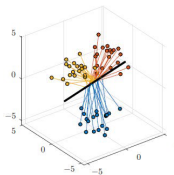
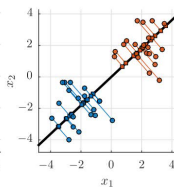
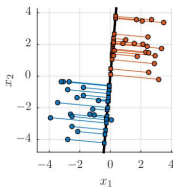
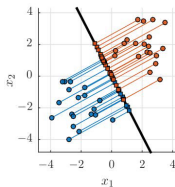
PCA for high-dimensional data

- Much data is high-dimensional
- We can project **high**-dimensional data to a **lower**-dimensional subspace.
- But what is a good projection?
- **Idea**: Select a projection that maximizes the variance of the projected data



PCA derivation

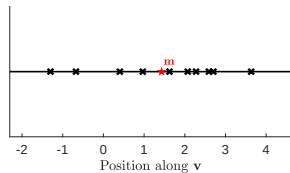
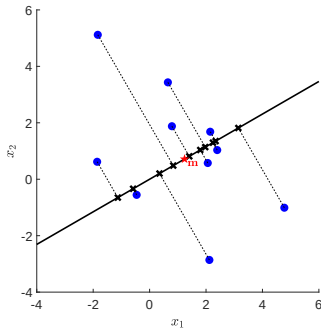
- Key idea: Find the direction in the M 'th dimensional space that captures the most variance.
- Approach:
 - 1 Measure variance when we project on a given direction given by a unit vector $\mathbf{v}$. The projection of a datapoint is $\mathbf{x}_i$, is $\mathbf{b}_i = \mathbf{x}_i^\top \mathbf{v}$.
 - 2 Find the direction $\mathbf{v}$ that maximizes the variance of the projected data points.



Determining the variance of a projection

$$b_i = \mathbf{x}_i^\top \mathbf{v}$$

$$\begin{aligned} \text{Var}[b] &= \frac{1}{N-1} \sum_{i=1}^N \left[b_i - \frac{1}{N} \sum_{j=1}^N b_j \right]^2 \\ &= \frac{1}{N-1} \sum_{i=1}^N \left[\mathbf{x}_i^\top \mathbf{v} - \frac{1}{N} \sum_{j=1}^N \mathbf{x}_j^\top \mathbf{v} \right]^2 \\ &= \frac{1}{N-1} \sum_{i=1}^N \left[\left(\mathbf{x}_i^\top - \frac{1}{N} \sum_{j=1}^N \mathbf{x}_j^\top \right) \mathbf{v} \right]^2 \\ &= \frac{1}{N-1} \sum_{i=1}^N (\tilde{\mathbf{x}}_i^\top \mathbf{v})^2 \quad \boxed{\tilde{\mathbf{x}}_i = \mathbf{x}_i - \mathbf{m}} \\ &= \frac{1}{N-1} \sum_{i=1}^N \mathbf{v}^\top \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top \mathbf{v} \\ &= \frac{1}{N-1} \mathbf{v}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{v} \end{aligned}$$



PCA derivation: Solving the optimization problem ✎

$$\arg \max_v \frac{1}{N-1} \mathbf{v}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{v} \quad s.t. \quad \|\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{v} = 1$$

- Use tricks from constrained optimization specifically to formulate "the Lagrangian". λ is the Lagrangian multiplier.

$$\arg \max_{\mathbf{v}, \lambda} = \frac{1}{N-1} \mathbf{v}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{v} + \lambda(1 - \mathbf{v}^\top \mathbf{v})$$

- The function we want to maximize is now:

$$\mathcal{L}(\mathbf{v}, \lambda) = \frac{1}{N-1} \mathbf{v}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{v} + \lambda(1 - \mathbf{v}^\top \mathbf{v})$$

- Taking the derivative and equating to 0:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{v}} = \frac{1}{N-1} 2 \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{v} - 2\lambda \mathbf{v} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 1 - \mathbf{v}^\top \mathbf{v} = 0 \quad (\text{not very informative?})$$

All-in-all we obtain:

$$\boxed{\frac{1}{N-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{v} = \lambda \mathbf{v}}$$

An eigenvalue problem of the form $A\mathbf{v} = \lambda\mathbf{v}$.

PCA derivation: **What about λ ?**

$$\frac{\partial \mathcal{L}}{\partial \mathbf{v}} = \frac{1}{N-1} 2 \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \mathbf{v} - 2\lambda \mathbf{v} = 0 \text{ or } \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \mathbf{v} = \lambda \mathbf{v}$$

$$\begin{aligned} \text{Var}[b] &= \frac{1}{N-1} \mathbf{v}^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \mathbf{v} \quad (\text{we know } \frac{1}{N-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \mathbf{v} = \lambda \mathbf{v}) \\ &= \mathbf{v}^T \lambda \mathbf{v} \\ &= \lambda \end{aligned}$$

To find the eigenvector (i.e. the principal component) of $\frac{1}{N-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$ that maximizes the variance, simply select the eigenvector with the largest corresponding eigenvalue λ .

PCA derivation: What is the matrix $\frac{1}{N-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$?

Recall the empirical co-variance:

$$\text{cov}[\mathbf{x}^{(l)}, \mathbf{x}^{(k)}] = \frac{1}{N-1} \sum_{i=1}^N (x_i^{(l)} - \hat{\mu}_{x^{(l)}})(x_i^{(k)} - \hat{\mu}_{x^{(k)}})$$

$$\hat{\mathbf{S}} = \begin{Bmatrix} 1 & 0.9 \\ 0.9 & 1.1 \end{Bmatrix}$$

And the covariance matrix:

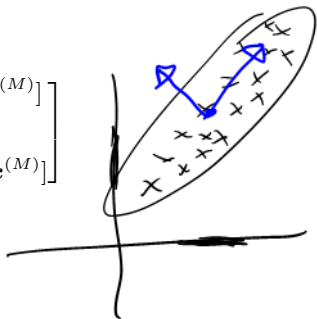
$$\hat{\mathbf{S}} = \begin{bmatrix} \text{cov}[\mathbf{x}^{(1)}, \mathbf{x}^{(1)}] & \dots & \text{cov}[\mathbf{x}^{(1)}, \mathbf{x}^{(M)}] \\ \vdots & & \vdots \\ \text{cov}[\mathbf{x}^{(M)}, \mathbf{x}^{(1)}] & \dots & \text{cov}[\mathbf{x}^{(M)}, \mathbf{x}^{(M)}] \end{bmatrix}$$

This can conveniently be computed as:

$$\hat{\mathbf{S}} = \frac{1}{N-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$$

where $\tilde{\mathbf{X}}$ is the centered data matrix as usual.

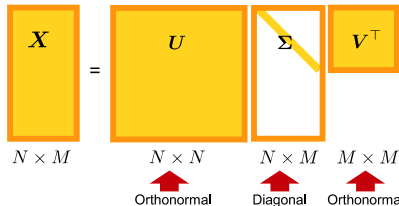
A strategy: Find the eigenvectors and eigenvalues of $\frac{1}{N-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$ and pick the eigenvector with the highest eigenvalue...



The Singular Value Decomposition (SVD)

Any matrix can be decomposed as follows:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$



$$\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_N], \quad \mathbf{V} = \begin{bmatrix} | & | & \dots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_M \\ | & | & & | \end{bmatrix}, \quad \mathbf{\Sigma} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_M \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

$$\mathbf{U}^\top \mathbf{U} = \mathbf{I}_{N \times N}, \quad \mathbf{V}^\top \mathbf{V} = \mathbf{I}_{M \times M},$$

$$Av = \lambda v$$

The Singular Value Decomposition (SVD)

Any matrix can be decomposed as follows (here the centered matrix):

$$\tilde{X} = U \Sigma V^T$$

$$\tilde{X} = U \Sigma V^T$$

$N \times M \quad N \times N \quad N \times M \quad M \times M$

$$\lambda_i = \frac{1}{N-1} \sigma_{ii}^2$$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_M \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

$$V^T = \begin{bmatrix} - & v_1 & - \\ - & v_2 & - \\ & \vdots & \\ - & v_M & - \end{bmatrix}$$

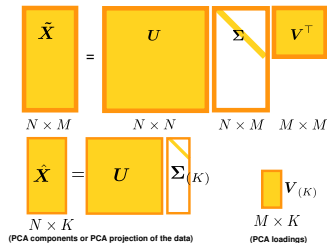
$$Ax = \lambda x$$

$$\begin{aligned} \frac{1}{N-1} (\tilde{X}^T \tilde{X}) v_i &= \frac{1}{N-1} (U \Sigma V^T)^T U \Sigma V^T v_i \\ &= \frac{1}{N-1} V \Sigma^T U^T U \Sigma V^T v_i \\ &= \frac{1}{N-1} V \Sigma^T \Sigma V^T v_i \\ &= \frac{1}{N-1} V \Sigma^T \Sigma e_i \\ &= \frac{1}{N-1} \sigma_{ii}^2 v_i \end{aligned}$$

Principal component analysis (PCA)

(Karl Pearson, 1901)

- 1 Subtract the column-wise mean, $\mathbf{m}$, from the data matrix $\mathbf{X}$, such that $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \mathbf{m}$
(we can standardize as well – consider the behavior)
- 2 Apply SVD to the centered matrix, i.e., $\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$
- 3 Select the K first columns of $\mathbf{V}$, $\mathbf{V}_{(K)} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_K]$, and first K columns of $\mathbf{\Sigma}$
- 4 *Optional:* Analyse the eigenvectors / principal components to see which of the original features contribute to the projection and how much.
- 5 *Optional:* Project the data onto the $\mathbf{V}_K$ basis and visualize the data in the new coordinate system.
- 6 *Optional:* Reconstruct any point in the new basis, $\mathbf{b}_i$ to the original, non-centered basis using $\mathbf{x}'_i = \mathbf{V}_{(K)}\mathbf{b}_i + \mathbf{m}$



PCA: What's really the role of Σ ?

$$\tilde{X} = U\Sigma V^\top$$

- Entries in the diagonal matrix Σ are called **singular values**
 - They are sorted (largest first)
 - Indicate how much variability is explained by the corresponding component
 - 1st component explains most of the variability
 - 2nd component explains most of the remaining variability
 - etc.

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_M \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} \quad (N \times M) \quad \sigma_1 \geq \sigma_2 \cdots \geq \sigma_M \geq 0$$

PCA: Explained Variance

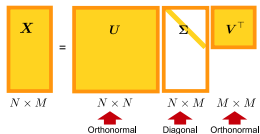
Recall that from SVD: $\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$

In the original but centered space, the coordinates of $\tilde{\mathbf{X}}$ project onto the first K components are:

$$\mathbf{X}' = \mathbf{U}\mathbf{\Sigma}_{(K)}\mathbf{V}_{(K)}^T$$

We can measure how much variance is retained in the reconstruction $\mathbf{X}'$:

$$\text{Explained var.} = \frac{\|\mathbf{X}'\|_F^2}{\|\tilde{\mathbf{X}}\|_F^2}$$



$$\text{cov}(\tilde{\mathbf{X}}) \propto \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$$

$$\text{trace}(\mathbf{AB}) = \text{trace}(\mathbf{BA})$$

$$\|\tilde{\mathbf{X}}\|_F^2 = \text{trace}(\tilde{\mathbf{X}}^T \tilde{\mathbf{X}}) = \text{trace}(\tilde{\mathbf{X}} \tilde{\mathbf{X}}^T)$$

PCA: Explained Variance

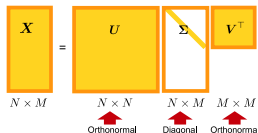
Recall that from SVD: $\tilde{X} = U\Sigma V^T$

In the original but centered space, the coordinates of $\tilde{X}$ project onto the first K components are:

$$X' = U\Sigma_{(K)}V_{(K)}^T$$

We can measure how much variance is retained in the reconstruction X' :

$$\text{Explained var.} = \frac{\|X'\|_F^2}{\|\tilde{X}\|_F^2} = \frac{\sum_{i=1}^K \sigma_i^2}{\sum_{i=1}^M \sigma_i^2}$$



$$\text{cov}(\tilde{X}) \propto \tilde{X}^T \tilde{X}$$

$$\text{trace}(AB) = \text{trace}(BA)$$

$$\begin{aligned}\|\tilde{X}\|_F^2 &= \text{trace}(\tilde{X}^T \tilde{X}) = \text{trace}(\tilde{X} \tilde{X}^T) \\ &= \text{trace}(U\Sigma V^T (U\Sigma V^T)^T) \\ &= \text{trace}(U\Sigma V^T V \Sigma^T U^T) \\ &= \text{trace}(U\Sigma \Sigma^T U^T) \\ &= \text{trace}(U^T U \Sigma \Sigma^T) \\ &= \text{trace}(\Sigma \Sigma^T) = \sum_i \sigma_i^2\end{aligned}$$

PCA: Explained Variance

Recall that from SVD: $\tilde{X} = U\Sigma V^T$

In the original but centered space, the coordinates of $\tilde{X}$ project onto the first K components are:

$$X' = U\Sigma_{(K)}V_{(K)}^T$$

We can measure how much variance is retained in the reconstruction X' :

$$\text{Explained var.} = \frac{\|X'\|_F^2}{\|\tilde{X}\|_F^2} = \frac{\sum_{i=1}^K \sigma_i^2}{\sum_{i=1}^M \sigma_i^2}$$

Similarly, the fraction of explained variance for the i 'th component is

$$\text{Explained var.} = \frac{\sigma_i^2}{\sum_{i=1}^M \sigma_i^2}$$

$$\text{var}[b_i] = \lambda_i$$

$$\lambda_i = \frac{\sigma_i^2}{N-1}$$

$$\begin{matrix} X & = & U & \Sigma & V^T \\ N \times M & & N \times N & N \times M & M \times M \\ & & \text{Orthonormal} & \text{Diagonal} & \text{Orthonormal} \end{matrix}$$

$$\frac{\lambda_1}{\sum \lambda_i}$$

$$\text{cov}(\tilde{X}) \propto \tilde{X}^T \tilde{X}$$

$$\text{trace}(AB) = \text{trace}(BA)$$

$$\begin{aligned} \|\tilde{X}\|_F^2 &= \text{trace}(\tilde{X}^T \tilde{X}) = \text{trace}(\tilde{X} \tilde{X}^T) \\ &= \text{trace}(U\Sigma V^T (U\Sigma V^T)^T) \\ &= \text{trace}(U\Sigma V^T V \Sigma^T U^T) \\ &= \text{trace}(U\Sigma \Sigma^T U^T) \\ &= \text{trace}(U^T U \Sigma \Sigma^T) \\ &= \text{trace}(\Sigma \Sigma^T) = \sum_i \sigma_i^2 \end{aligned}$$

Quiz 1: PCA

$$\|X\|_F^2 = \sum_{i=1}^M \sigma_i^2$$

$$\text{Exp var.} = \frac{\sum_{i=1}^M \sigma_i^2}{M}$$

No.	Attribute description	Abbrev.
x_1	Age (in years)	AGE
x_2	Gender (Female=0, Male=1)	GDR
x_3	Total Bilirubin	TB
x_4	Direct Bilirubin	DB
x_5	Alkaline Phosphatase	AP
x_6	Alamine Aminotransferase	AlA
x_7	Aspartate Aminotransferase	AsA
x_8	Total Protiens	TP
x_9	Albumin	AB
x_{10}	Albumin to Globulin ratio	A/G
y	0=No liver disease, 1=Liver disease	LD

Table 1: Attributes in a study on liver disease among Indians living in the north eastern part of Andhra Pradesh, India. (taken from <http://archive.ics.uci.edu/ml/datasets/ILPD> +%28Indian+Liver+Patient+Dataset%29). The data has 10 input attributes x_1-x_{10} and one output variable y which defines whether the subject considered has a liver disease ($y = 1$) or not ($y = 0$). x_3-x_9 are non-negative measurements giving the concentrations of various quantities measured in a blood test. x_{10} gives the ratio of Albumin to Globulin in the blood.

A PCA analysis is applied to the standardized data based on the attributes x_1-x_{10} . The squared Frobenius norm of the standardized data matrix X is given by $\|X\|_F^2 = 5780.0$. The first four singular values are $\sigma_1 = 40.1$, $\sigma_2 = 34.2$, $\sigma_3 = 28.1$, and $\sigma_4 = 24.8$, Which of the following statements is correct?

- The first PCA component accounts for more than 35 % of the variation. $\frac{40.1^2}{5780} = 27.8\%$
- The second PCA component accounts for more than 30 % of the variation. $\frac{34.2^2}{5780} = 20\%$
- The first three PCA components account for less than 70 % of the variation in the data.
- The fourth PCA component accounts for less than 10 % of the variation in the data.
- Don't know. $\frac{24.8^2}{5780} = 10.6\%$

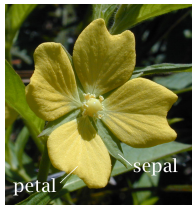
$$\frac{40.1^2 + 34.2^2 + 28.1^2}{5780} = 61.7\%$$

Solution:

The i^{th} principal component accounts for $\frac{\sigma_i^2}{\sum_j \sigma_j^2} = \frac{\sigma_i^2}{\|X\|_F^2}$. We therefore have that the first PCA component accounts for $\frac{40.1^2}{5780.0} = 27.8\%$, the second $\frac{34.2^2}{5780.0} = 20.2\%$, and the first three principal components account for $\frac{40.1^2 + 34.2^2 + 28.1^2}{5780.0} = 61.7\%$ of the variation whereas the fourth principal component accounts for $\frac{24.8^2}{5780.0} = 10.6\%$. Thus, the first three PCA components account for less than 70% of the variation in the data.

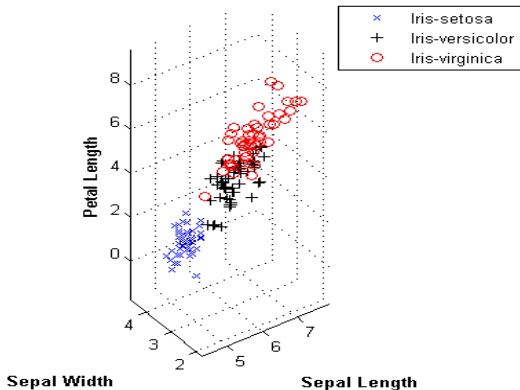
Example: PCA on Fisher's Iris Data

- We will presently consider the first 3 attributes, i.e. Sepal length, Sepal Width and Petal Length.
- Three types of flowers: Iris Setosa, Iris Versicolor, Iris Virginica



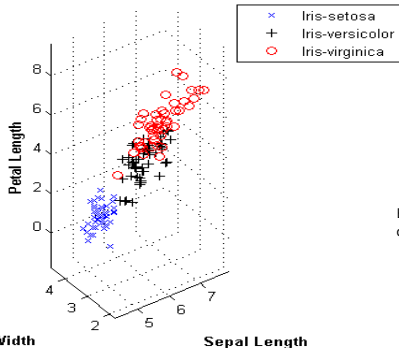
Flower ID	Attribute			
	Sepal Length	Sepal Width	Petal Length	Petal Width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
4	4.6	3.1	1.5	0.2
.	.	.	.	.
.	.	.	.	.
150	5.9	3.0	5.1	1.8

3D scatter plot of Iris Data



What fraction of the total variation in the data
will the first principal component account for?

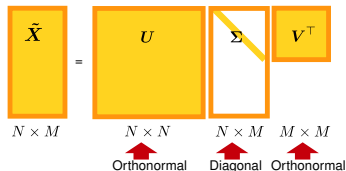
3D scatter plot of Iris Data



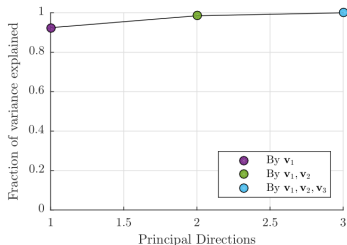
What fraction of the total variation in the data will the first principal component account for?

- 1) Subtract the mean
- 2) Apply singular value decomposition (SVD)

$$\tilde{X} = U \Sigma V^T$$

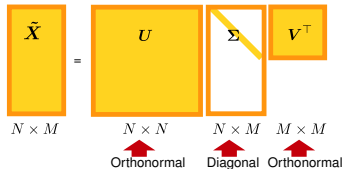


Evaluate the singular values to determine how much of the dynamics is lost when reducing the dimensionality



Visualization of the PCA projections of the data

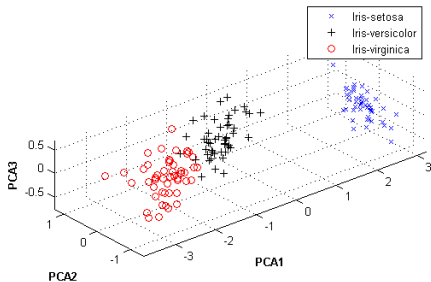
$$\tilde{X} = U \Sigma V^T$$



$$PCA1 : b_1 = \tilde{X} v_1 = u_1 \sigma_1$$

$$PCA2 : b_2 = \tilde{X} v_2 = u_2 \sigma_2$$

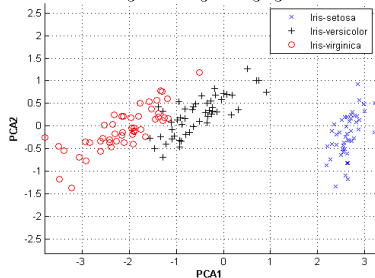
$$PCA3 : b_3 = \tilde{X} v_3 = u_3 \sigma_3$$



The principal directions V

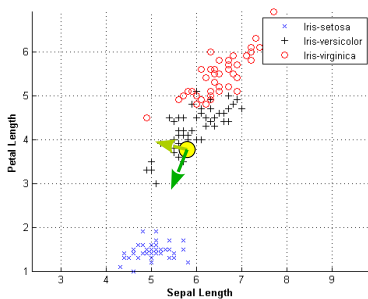
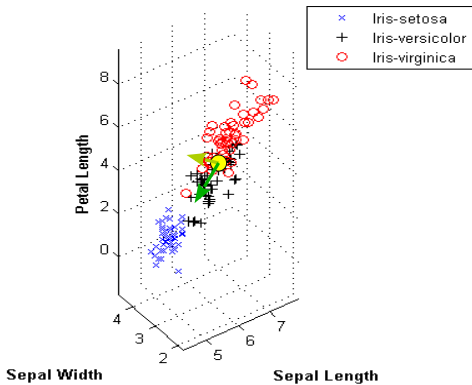
Sepal Length
 Sepal Width
 Petal Length

$$V = \begin{bmatrix} -0.39 & -0.64 & -0.66 \\ 0.09 & -0.74 & 0.66 \\ -0.92 & 0.20 & 0.35 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 5.8 \\ 3.1 \\ 3.8 \end{bmatrix}, \quad \vec{v}_1 = \begin{bmatrix} -0.39 \\ 0.09 \\ -0.92 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} -0.64 \\ -0.74 \\ 0.20 \end{bmatrix}$$

Sepal Length
 Sepal Width
 Petal Length



Quiz2: PCA

No.	Attribute description	Abbrev.
x_1	Age (in years)	AGE
x_2	Gender (Female==0, Male==1)	GDR
x_3	Total Bilirubin	TB
x_4	Direct Bilirubin	DB
x_5	Alkaline Phosphatase	AP
x_6	Alamine Aminotransferase	AlA
x_7	Aspartate Aminotransferase	AsA
x_8	Total Proteins	TP
x_9	Albumin	AB
x_{10}	Albumin to Globulin ratio	A/G
y	0=No liver disease, 1=Liver disease	LD

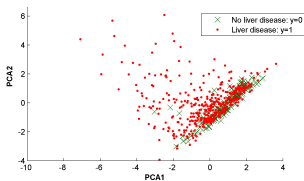


Figure 1: Principal component 1 (PCA1) plotted against principal component 2 (PCA2).

The first and second principal component directions

of the liver-dataset are

$$v_1 = \begin{bmatrix} -0.1404 \\ -0.1090 \\ -0.4115 \\ -0.4179 \\ -0.2468 \\ -0.2682 \\ -0.3009 \\ 0.2781 \\ 0.4375 \\ 0.3638 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -0.2859 \\ 0.0130 \\ 0.2510 \\ 0.2622 \\ 0.0525 \\ 0.4162 \\ 0.3927 \\ 0.4197 \\ 0.4323 \\ 0.3052 \end{bmatrix}.$$

In the figure, the data projected onto the first two principal components is plotted, and the colors indicate the presence of liver disease. Which of the following statements is *correct*?

- Relatively high values of AGE, GDR, TB, DB, AP, AlA, and AsA and low values of TP, AB, and A/G will result in a positive projection onto the first principal component.
- Relatively low values of the projection onto PCA1 and high values of the projection onto PCA2 indicates the subject does not have a liver disease.
- PCA2 mainly discriminate between old subjects with low measurements of TB, DB, AlA, AsA, TP, AB, and A/G from young subjects with high values of TB, DB, AlA, AsA, TP, AB, and A/G.
- The principal component directions are not guaranteed to be orthogonal to each other since the data has been standardized.
- Don't know.

Solution:

AGE, GDR, TB, DB, AP, AlA, and AsA have negative coefficients of PCA1 whereas TP, AB, and A/G have positive coefficients resulting in a negative projection onto the first principal component, thus this is correct. From the figure we observe that observations with low values of PCA1 and high values of PCA2 in general have a red dot meaning they have a liver disease. For PCA2 we observe that AGE has a negative value whereas the remaining entities

have positive values while GDR and AP have small amplitudes. As a result PCA2 mainly discriminate between young subjects with high measurements of TD, DB, AlA, AsA, TP, AB, and A/G from old subjects with low values of TD, DB, AlA, AsA, TP, AB, and A/G hence this is correct. The principal component directions are always orthogonal to each other irrespective of the data preprocessing.

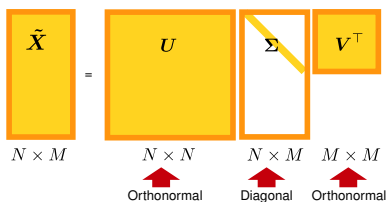
Example: Visualization of hand written digits

- Data matrix

$$X = \begin{bmatrix} \text{---} & x_1 & \text{---} \\ \text{---} & x_2 & \text{---} \\ & \vdots & \\ \text{---} & x_N & \text{---} \end{bmatrix}$$


If each image is 28 x 28 pixels then X is a $N \times 784$ matrix

- Principal component analysis

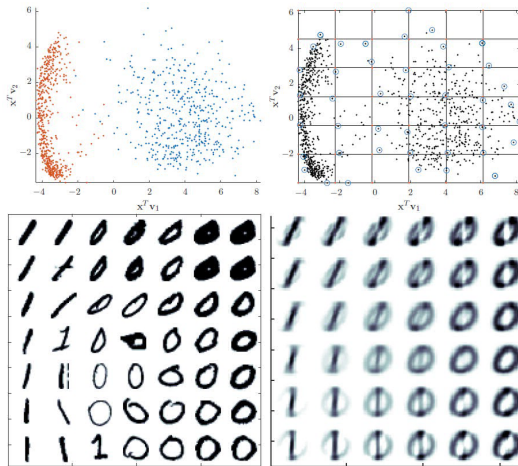
$$\tilde{X} = U \Sigma V^T$$


$\tilde{X}$ U Σ V^T

$N \times M$ $N \times N$ $N \times M$ $M \times M$

Orthonormal Diagonal Orthonormal

Visualization of hand written digits



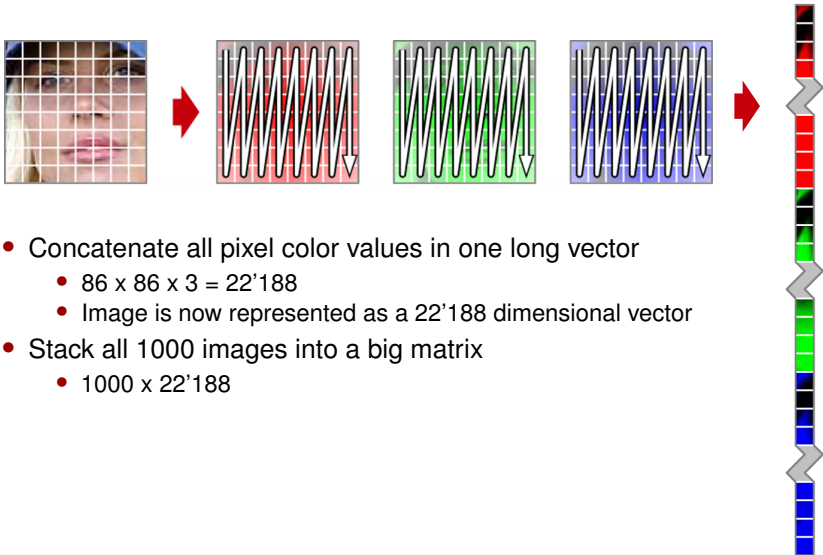
Example: Principal component analysis of faces

- 1000 images, 86 x 86 pixels, 3 RGB intensities

Tamara Berg: “Faces in the wild”

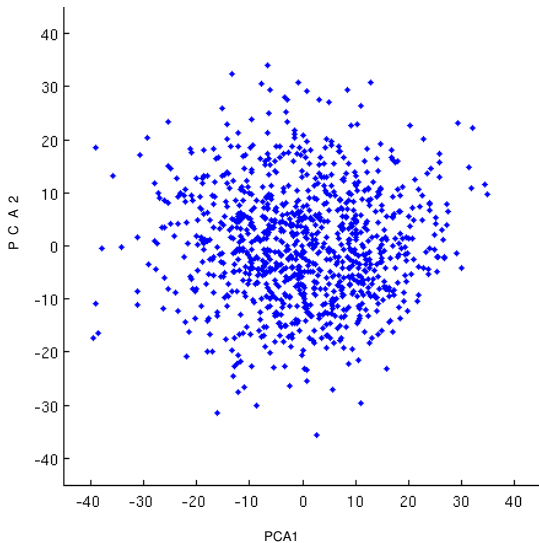


Pre-processing

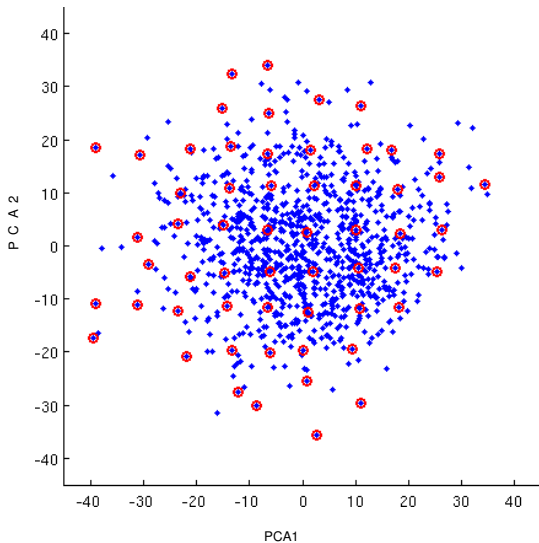


- Concatenate all pixel color values in one long vector
 - $86 \times 86 \times 3 = 22'188$
 - Image is now represented as a 22'188 dimensional vector
- Stack all 1000 images into a big matrix
 - $1000 \times 22'188$

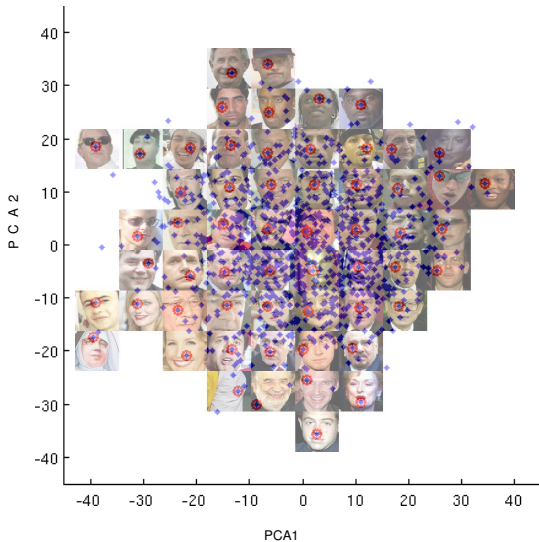
PCA on face images



PCA on face images

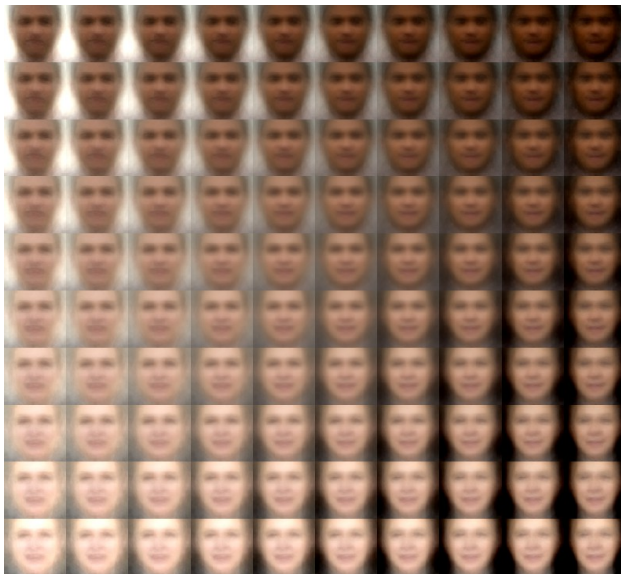


PCA on face images



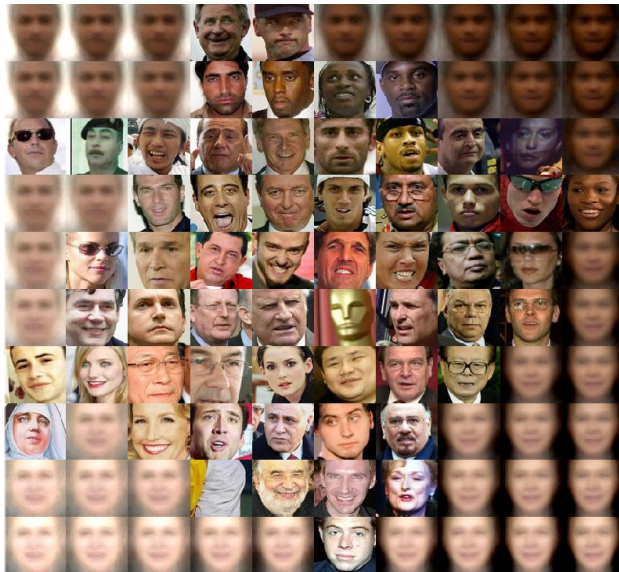


- What information do the two principal axes capture?





What information do the two principal axes capture?



Example: PCA as compression

- Only include a few components: $\hat{x} = V_{(K)}b + m \quad K = \{2, 5, 20, 50, 100\}$



Data and domain driven feature extraction

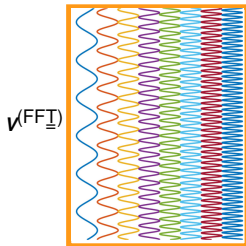
PCA is an example of a data driven approach for feature extraction.

i.e., we define from data the features extracted in terms of the projections $\mathbf{V}^{(PCA)}$ that preserve most of the variance in the data.

$$\begin{array}{c} \tilde{\mathbf{X}} \\ N \times M \end{array} = \begin{array}{c} \mathbf{U} \\ N \times N \end{array} \begin{array}{c} \Sigma \\ N \times M \end{array} \begin{array}{c} \mathbf{V}^T \\ M \times M \end{array}$$

The fourier transform is an example of a domain driven approach for feature extraction

i.e., in the analysis of sound good features are often to use spectral representations. These can be found by projecting the data using the so-called fourier transform matrix $\mathbf{V}^{(FFT)}$ where the components are defined as specific frequencies such that the projection of the data onto these frequencies defines the extent to which these frequencies are present in the data.



Resources

DTU Compute Our online PCA demo which highlights key concepts of PCA such as the effect of normalization, variance explained, and much more (<http://www2.imm.dtu.dk/courses/02450/DemoPCA.html>)

<https://arxiv.org> A great and more in-depth tutorial on PCA
(<https://arxiv.org/abs/1404.1100>)

<https://www.3blue1brown.com> An great, animated recap of linear algebra
(<https://www.3blue1brown.com/essence-of-linear-algebra-page/>)